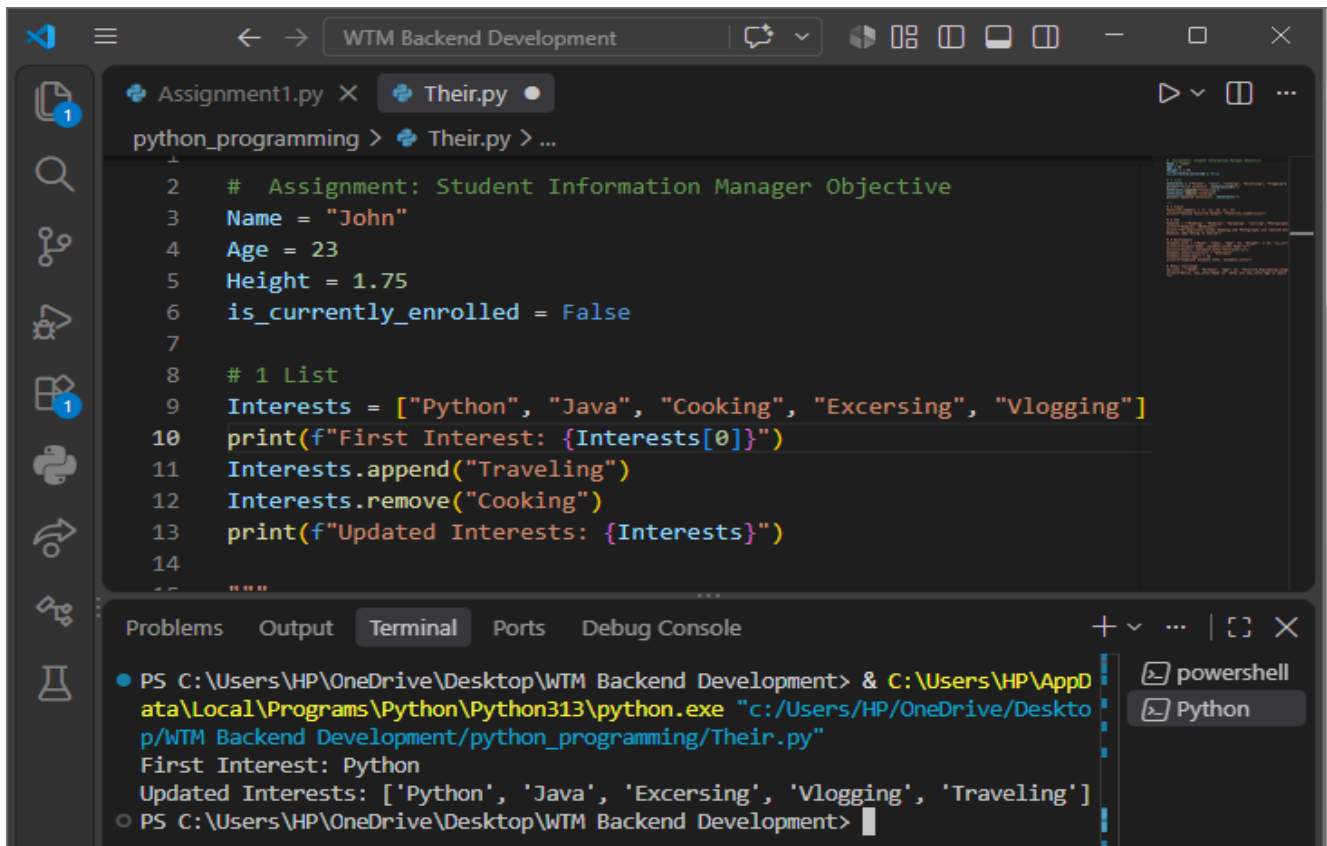


1. List



The screenshot shows the Visual Studio Code editor with a file named 'Their.py' open. The code defines a student's information and a list of interests. The terminal output shows the execution of the script, which prints the first interest and the updated list after adding 'Traveling' and removing 'Cooking'.

```
python_programming > Their.py > ...  
1  
2 # Assignment: Student Information Manager Objective  
3 Name = "John"  
4 Age = 23  
5 Height = 1.75  
6 is_currently_enrolled = False  
7  
8 # 1 List  
9 Interests = ["Python", "Java", "Cooking", "Excercise", "Vlogging"]  
10 print(f"First Interest: {Interests[0]}")  
11 Interests.append("Traveling")  
12 Interests.remove("Cooking")  
13 print(f"Updated Interests: {Interests}")  
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```

3. Set

```
19
20 # 3 Set
21 Hobbies = {"Reading", "Reading", "Swimming",
22           "Cycling", "Photography", "Hiking", "Photography"}
23 print(f"Hobbies: {Hobbies}")
24 print("The duplicate values Reading and Photography are " \
25       "removed because sets only allow unique elements.")
26 Hobbies.add("Going to Church")
27
```

Problems Output Terminal Ports Debug Console

```
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> ^C
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> & C:\Users\HP\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/HP/OneDrive/Desktop/WTM Backend Development/python_programming/Their.py"
Hobbies: {'Reading', 'Swimming', 'Hiking', 'Photography', 'Cycling'}
The duplicate values Reading and Photography are removed because sets only allow unique elements.
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> 
```

4. Dictionary

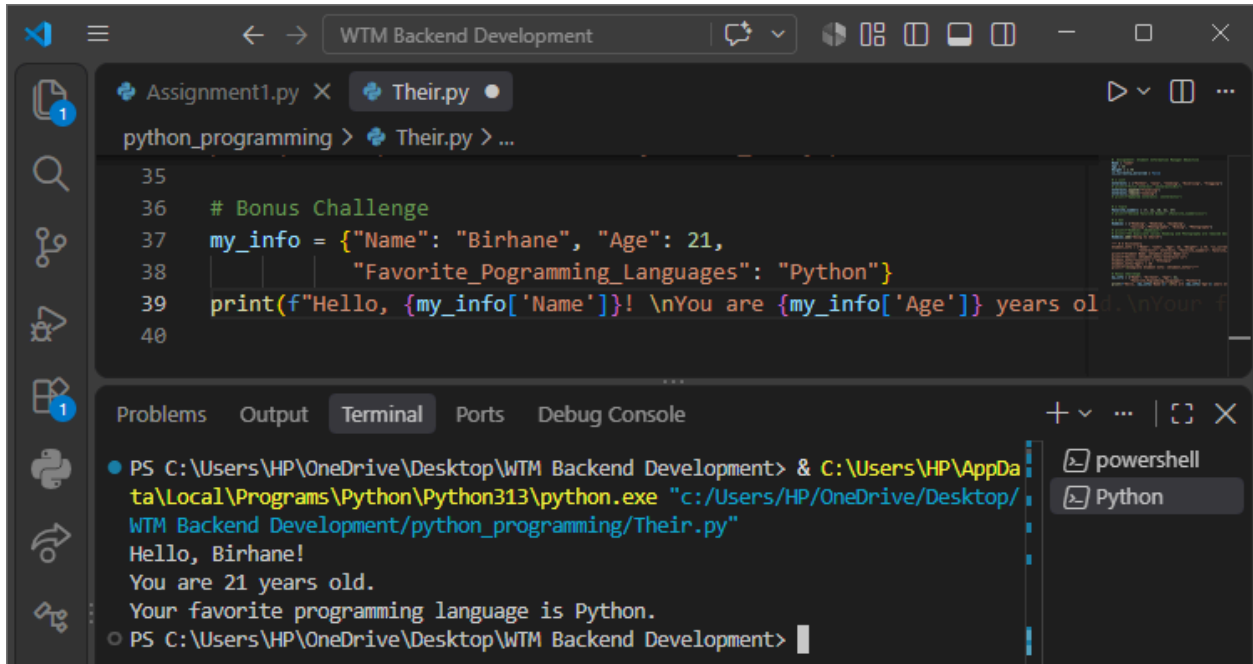
```
27 # 4 Dictionary
28 Student_Info = {"Name": "John", "Age": 23, "Height": 1.75, "is_currently_enrolled": False,
29               "Interests": Interests, "Favorite_numbers": Favorite_numbers, "Hobbies": Hobbies}
30 print(f"Student Name: {Student_Info['Name']}")
31 print(f"Skills: {Student_Info['Interests']}")
32 Student_Info["Country"] = "Ethiopia"
33 Student_Info["Age"] = 28
34 print(f"\nComplete Student Info: {Student_Info}")
35
```

Problems Output Terminal Ports Debug Console

```
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> & C:\Users\HP\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/HP/OneDrive/Desktop/WTM Backend Development/python_programming/Their.py"
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development>
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> ^C
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> & C:\Users\HP\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/HP/OneDrive/Desktop/WTM Backend Development/python_programming/Their.py"
Student Name: John
Skills: ['Python', 'Java', 'Excercise', 'Vlogging', 'Traveling']

Complete Student Info: {'Name': 'John', 'Age': 28, 'Height': 1.75, 'is_currently_enrolled': False, 'Interests': ['Python', 'Java', 'Excercise', 'Vlogging', 'Traveling'], 'Favorite_numbers': (7, 12, 16, 21, 27), 'Hobbies': {'Hiking', 'Photography', 'Going to Church', 'Swimming', 'Cycling', 'Reading'}, 'Country': 'Ethiopia'}
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> 
```

#Bonus Challenge



The screenshot shows a Visual Studio Code window titled "WTM Backend Development". The editor has two tabs: "Assignment1.py" and "Their.py". The "Their.py" tab is active, showing a Python script with the following code:

```
35
36 # Bonus Challenge
37 my_info = {"Name": "Birhane", "Age": 21,
38            "Favorite_Pogramming_Languages": "Python"}
39 print(f"Hello, {my_info['Name']}! \nYou are {my_info['Age']} years ol
40
```

The bottom panel shows the "Terminal" tab with the following output:

```
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development> & C:\Users\HP\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/HP/OneDrive/Desktop/WTM Backend Development/python_programming/Their.py"
Hello, Birhane!
You are 21 years old.
Your favorite programming language is Python.
PS C:\Users\HP\OneDrive\Desktop\WTM Backend Development>
```

A short explanation of the difference between:

- ❖ **List:** An ordered and changeable collection. It allows duplicate values.
Example: ["Python", "Java", "Python"]
- ❖ **Tuple:** An ordered collection that cannot be changed after it is created. It allows duplicates.
Example: ("Python", "Java", "Python")
- ❖ **Set:** An unordered collection that does not allow duplicate values.
Example: {"Python", "Java"}
- ❖ **Dictionary:** A collection of key-value pairs, where each key is used to access its corresponding value.
Example: {"Name": "Abebe", "Age": 20}

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